# Performance Analysis of RIS-Assisted ISAC-NOMA Uplink System Under Imperfect SIC

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***Abstract*—This paper investigates the performance of a reconfigurable intelligent surface (RIS)-assisted non-orthogonal multiple access (NOMA) integrated sensing and communication (ISAC) system comprising two communication users and one target. An RIS partitioning strategy is used, wherein separate RIS zones are allocated to support communication users and radar echo reflection simultaneously. The practical effect of inter-zone interference among RIS partitions is considered and the probability density function of the effective channel power gain from communication users to the base station considering reflections from all RIS zones are provided. Under the imperfect successive interference cancellation scenario, closed-form expressions for the outage probability of communication users and sensing target are derived. We also obtain analytical expression of average sensing information rate that characterizes long-term sensing throughput. Numerical results reveal that our proposed scheme substantially outperforms orthogonal multiple access scheme and bring out the efficacy of deploying RIS and partitioning coefficient configurations for reliable communication and sensing in ISAC-NOMA system.**

***Keywords*—Reconfigurable intelligent surface, non-orthogonal multiple access, integrated sensing and communication, outage probability, average sensing information rate, imperfect successive interference cancellation.**

## I. Introduction

### A. Background and Motivation

Integrated sensing and communication (ISAC) has emerged as a key enabler for future sixth generation wireless networks, offering the capability to unify wireless communication and environmental sensing within a shared infrastructure [1]. By jointly optimizing spectrum, hardware, and signal processing, ISAC finds applications in internet of things (IoT), vehicle-to-everything communications, and connected autonomous systems [2]. When multiple communication users are present, non-orthogonal multiple access (NOMA) is a promising solution to serve users by sharing the same time-frequency resources and multiplexing them in the power domain employing successive interference cancellation (SIC) to mitigate interference [3].

However, the ISAC-NOMA system's performance can be limited due to large path loss and shadowing effect. Reconfigurable intelligent surface (RIS) offers a low-cost and energy-efficient solution by reconfiguring the wireless environment by its passive beamforming technique [4]. By creating controllable virtual line-of-sight (LoS) links, RIS enhances received signal strength, sensing accuracy, and communication users' performance in obstructed scenarios [5].

### B. Related Works for RIS-Assisted ISAC-NOMA System

In [6], a dual-functional base station (BS) with joint active and passive beamforming technique was proposed to maximize the RIS minimum beampattern gain. The work in [7] developed a hybrid orthogonal multiple access (OMA)–NOMA model and an alternating optimization algorithm was proposed to maximize the minimum beampattern gain. In [8], closed-form outage and detection probabilities were derived in high signal-to-noise ratio (SNR) regime considering the Rayleigh-Rician cascaded channel, where the RIS supports only the far user. The study in [9] analyzed communication outage probability and sensing rate in a multi-sector RIS framework under Nakagami-m fading. In [10] communication-outage probability considering imperfect SIC (impSIC) was analyzed in a satellite-based RIS-ISAC-NOMA system under shadowed-rician and Nakagami-m fading models.

TABLE I: Comparative Study of Prior Works

| Ref. | UL/DL | impSIC | RIS | Analysis |
|---|---|---|---|---|
| [6] | DL | ✗ | Phase optimized | No |
| [7] | DL | ✗ | Phase optimized | No |
| [8] | DL | ✗ | Phase optimized | OP-C, PoD |
| [9] | DL | ✗ | Partitioned and optimized | OP-C, sensing rate |
| [10] | DL | ✓ | Optimized | OP-C |
| **Proposed** | **UL** | ✓ | **Partitioned, optimized + I** | **OP-C, OP-S, ASIR** |

I = Inter-zone interference; OP-C = Communication outage probability; OP-S = Sensing outage probability; PoD = Probability of detection.

### C. Motivations and Contributions

From Table I, we note that [7]–[10] assume perfect SIC and focus only on downlink (DL) communication. No performance analysis is provided in [6], [7]. The work in [8]–[10] analyze outage probability only for communication task and no sensing metric is analyzed in [10]. Further, [9] do not model inter-sector interference, which arises due to reflections of the transmitted signal from the BS via different RIS sectors.

Motivated by this gap, our contributions are as follows. (i) *Model:* We investigate into an RIS-assisted *uplink* ISAC-NOMA system comprising two communication users and a sensing target. The RIS is partitioned into three zones, which are phase-optimized to reflect the near and far users signals and target echo, respectively. The impact of communication signal-echo interference, RIS inter-zone reflections, and imp-SIC are all modeled. (ii) *Analysis:* We derive probability density function (PDF) of the effective channel power gains from the users to the BS and for radar echo considering reflections

from all RIS zones. Analytical expressions for the outage probability of both communication and sensing are provided under Nakagami-m fading. We further analyze an average sensing information rate (ASIR). (iii) *Results:* Simulation results demonstrate the impact of impSIC factor, transmit powers, no. of RIS reflective elements, target location, and choice of RIS partitioning coefficients on communication and sensing performance, and benchmark NOMA with OMA.

## II. System Model

As shown in Fig. 1, we consider an RIS-assisted uplink ISAC-NOMA system consists of a BS, two communication users – a near user $D_n$, a far user $D_f$ and a sensing target $T$. Each node is equipped with a single antenna, and the BS operates as a dual-functional unit supporting simultaneous communication and sensing functionalities. Owing to obstacles, there are no direct LoS links between the communication users and BS, and BS and target [6]. Thus, the users' signals and the target echo are reflected to the BS via an RIS ($R$) having $L$ passive reflecting elements (REs).

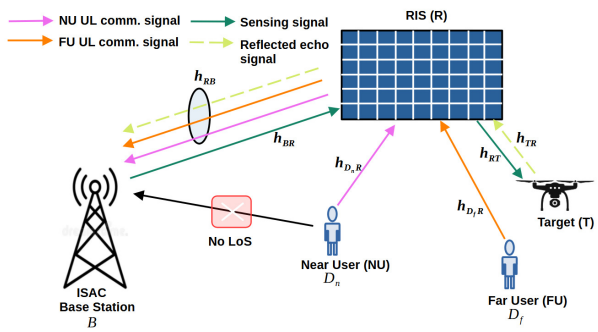



Fig. 1: An RIS-assisted ISAC-NOMA uplink system model.

*Channel Model:* The complex baseband channel gain vector from $D_n$ to $R$ is $\mathbf{h}_{D_nR}$, from $D_f$ to $R$ is $\mathbf{h}_{D_fR}$, from $T$ to $R$ is $\mathbf{h}_{TR}$, from $R$ to $T$ is $\mathbf{h}_{RT}$, from BS to $R$ is $\mathbf{h}_{BR}$, and from $R$ to BS is $\mathbf{h}_{RB} \in \mathbb{C}^{L\times 1}$. We consider independent quasi-static Nakagami-$m$ fading model between the two nodes $X$ and $Y$, i.e., $|h_{XY}| \sim \text{Nakagami}(m_{XY}, \Omega_{XY})$, where $m_{XY}$ and $\Omega_{XY}$ denote the shape and spread parameters, respectively [11, (9)]. Further, $\Omega_{XY} = \mathbb{E}\left[|h_{XY}|^2\right] = Cd_{XY}^{-\alpha}$, where $d_{XY}$ is the distance between the two nodes $X$ and $Y$, $\alpha$ is the path loss exponent, and $C = c/(4\pi f_c)^2$ computed at reference distance $d_0 = 1\,\text{m}$ with the speed of light $c = 3\times 10^8$m/s, and the carrier frequency $f_c$.

*RIS Partitioning:* The RIS is partitioned into three zones, $\mathcal{Z}_j$, where $j \in \{n, f, T\}$, corresponding to the near user, far user, and the sensing target, respectively. The proportions of REs allocated to these zones are determined by the partitioning coefficients $a_n$, $a_f$, and $a_T$ satisfying $a_n + a_f + a_T = 1$. Accordingly, the number of REs allocated in each zone is denoted by $|\mathcal{Z}_j| = L_j$, where $L_j = a_jL$. Each RIS zone $\mathcal{Z}_j$ is associated with a diagonal reflection matrix $\mathbf{\Theta}_j = \text{diag}(\eta_{l_j}e^{j\theta_{l_j}}) \in \mathbb{C}^{L_j\times L_j}$, where $\eta_{l_j} \in [0,1]$ and $\theta_{l_j} \in [0, 2\pi)$ are the reflection amplitude and the phase shift of the $l_j^{th}$ RE, respectively, for $l_j \in Z_j$. To maximize the reflection power of the RIS, we assume unity reflection amplitude, as in [12].

Let $\theta_{XY}$ denote the phases of the corresponding baseband channel gains between any two nodes $X$ and $Y$. For the near-user partition, the phase shift of each RE is set as $\theta_{l_n} = -(\theta_{D_nR_{l_n}} + \theta_{R_{l_n}B})$, $l_n \in Z_n$ in order to maximize the received signal power from $D_n$ at the BS. Similarly, for the far-user partition, the phase shift is configured as $\theta_{l_f} = -(\theta_{D_fR_{l_f}} + \theta_{R_{l_f}B})$, $l_f \in Z_f$. For the target-reflecting zone, the phase shift of each RE is set as $\theta_{l_T} = -(\theta_{TR_{l_T}} + \theta_{R_{l_T}B})$, $l_T \in Z_T$, aiming to coherently combine the reflected radar echoes at the BS for improved sensing performance. Altogether, $\mathbf{\Theta} = \text{blkdiag}(\mathbf{\Theta}_n, \mathbf{\Theta}_f, \mathbf{\Theta}_T)$ represents the overall RIS reflection matrix of $L$ REs.

### A. Signal Model and SINRs Under Imperfect SIC

The message signals transmitted by $D_n$ and $D_f$ are denoted by $x_n(t)$ and $x_f(t)$, respectively, while the reflected sensing signal from the target is denoted by $x_s(t-\tau)$, where $\tau$ is the reflection delay and $x_s(t)$ is the sensing signal transmitted from the BS with all having unit energy. The transmit powers of $D_n$, $D_f$, and the BS are denoted as $P_n$, $P_f$, and $P_T$, respectively. The BS sensing signal reaches $T$ via the RIS and is reflected back through the same path. To ensure distinct received power levels between users, a power back-off strategy similar to [13] is employed. Specifically, $P_f = a_1P_n$, where $a_1 = 10^{-P_\Delta/10}$, i.e., $P_n$ is $P_\Delta$ (dB) stronger than $P_f$.

The RIS-reflected effective baseband channel gain from $D_n$ to the BS is $h_{D_n} = \mathbf{h}_{RB}^T\mathbf{\Theta}\mathbf{h}_{D_nR}$, from $D_f$ to the BS is $h_{D_f} = \mathbf{h}_{RB}^T\mathbf{\Theta}\mathbf{h}_{D_fR}$, from the BS to $T$ is $h_T' = \mathbf{h}_{RT}^T\mathbf{\Theta}\mathbf{h}_{BR}$, and from $T$ to the BS is $h_T = \mathbf{h}_{RB}^T\mathbf{\Theta}\mathbf{h}_{TR} \in \mathbb{C}^{1\times 1}$. Further, $n_{BS}(t) \sim \mathcal{CN}(0, \sigma_{BS}^2)$ denotes the complex additive white Gaussian noise (AWGN) with covariance $\sigma_{BS}^2$ at the BS receiver. Then, the composite received signal at the BS can be expressed as

$$y_{BS}(t) = \sqrt{P_n}\,h_{D_n}x_n(t) + \sqrt{P_f}\,h_{D_f}x_f(t) + \sqrt{P_T}\sqrt{\varrho}\,h_T'h_Tx_s(t-\tau) + n_{BS}(t). \quad (1)$$

where $\varrho = C_r\gamma^2B^2\sigma_\tau^2$, with $C_r = \sigma_{\text{RCS}}\,\lambda^2/(4\pi)^3$, $\lambda$ is the wavelength of the carrier, $\sigma_{RCS}$ denotes the radar cross section, $B$ is the bandwidth, $\gamma^2 = 2\pi^2/12$, and $\sigma_\tau^2$ denotes the variance of the time-delay fluctuation $\tau$ [14].[1]

*Signal Decoding Order:* The radar echo experiences double attenuation over the RIS-assisted forward and backward paths, whereas the uplink communication signals traverse a single effective channel via RIS. Therefore, the received power of the communication signals is more than that of the sensing echo, i.e., $P_n|h_{D_n}|^2 > P_f|h_{D_f}|^2 > \varrho P_T|g_T|^2$, where $g_T = h_T'h_T$ represents the effective baseband channel gain from BS-to-$T$-to-BS via RIS for the radar echo. The BS employs SIC to jointly detect the communication and radar signals and mitigate co-channel interference. In the SIC decoding order, the near-user signal is decoded first, as large-scale path loss dominates the received signal strength, followed by decoding the far user signal and finally, decoding radar echo.

[1] $\varrho$ represents the radar echo power scaling factor that aggregates the effects of radar cross section, signal bandwidth, and time-delay fluctuation variance.

*Signal-to-Interference-plus-Noise Ratio (SINR):* The SINR for detecting the near user's signal at the BS is given by

$$\gamma_{D_n} = \frac{|h_{D_n}^{\mathcal{Z}_n}|^2 P_n}{|h_{D_f}|^2 P_f + \varrho |g_T|^2 P_T + \sigma_{BS}^2}, \tag{2}$$

where the received near-user signal power arises from the reflections contributed by the RIS zone, $\mathcal{Z}_n$. This zone is phase-optimized to maximize the received signal from $D_n$ at the BS. Accordingly, the effective cascaded channel corresponding to the RIS zone $\mathcal{Z}_n$ is expressed as $h_{D_n}^{\mathcal{Z}_n} = \left(\mathbf{h}_{RB}^{\mathcal{Z}_n}\right)^T \mathbf{\Theta}_n \mathbf{h}_{D_n R}^{\mathcal{Z}_n}$, where $\mathbf{h}_{D_n R}^{\mathcal{Z}_n}$ and $\mathbf{h}_{RB}^{\mathcal{Z}_n} \in \mathbb{C}^{L_n \times 1}$ denote the $D_n$-to-RIS and RIS-to-BS channel vectors corresponding to zone $\mathcal{Z}_n$, respectively.

By the SIC process, the decoded near-user's signal is subtracted from the composite received signal. Under imperfect SIC, the received SINR for detecting far user's signal $x_f(t)$ at the BS is given by

$$\gamma_{D_f} = \frac{|h_{D_f}^{\mathcal{Z}_f}|^2 P_f}{\delta_n |h_{D_n}|^2 P_n + \varrho |g_T|^2 P_T + \sigma_{BS}^2}, \tag{3}$$

where $0 < \delta_n < 1$ represents the impSIC factor of residual received signal power from $D_n$; $\delta_n = 0$ corresponds to the perfect SIC and $h_{D_f}^{\mathcal{Z}_f} = \left(\mathbf{h}_{RB}^{\mathcal{Z}_f}\right)^T \mathbf{\Theta_f} \mathbf{h}_{D_f R}^{\mathcal{Z}_f}$ represents the effective cascaded channel from $D_f$-to-BS via RIS zone $\mathcal{Z}_f$, where $\mathbf{h}_{D_f R}^{\mathcal{Z}_f}$ and $\mathbf{h}_{RB}^{\mathcal{Z}_f} \in \mathbb{C}^{L_f \times 1}$ denote $D_f$-to-RIS and RIS-to-BS channel vectors corresponding to zone $\mathcal{Z}_f$, respectively. Finally, the SINR to detect the radar echo at the BS under imperfect SIC is given by

$$\gamma_{BS} = \frac{\varrho |g_T^{\mathcal{Z}_T}|^2 P_T}{\delta_n |h_{D_n}|^2 P_n + \delta_f |h_{D_f}|^2 P_f + \sigma_{BS}^2}, \tag{4}$$

where $0 < \delta_f < 1$ represents the impSIC factor of the residual received signal power from $D_f$ and $g_T^{\mathcal{Z}_T} = \left(\mathbf{h}_{RT}^{\mathcal{Z}_T}\right)^T \mathbf{\Theta}_T \mathbf{h}_{BR}^{\mathcal{Z}_T} \left(\mathbf{h}_{RB}^{\mathcal{Z}_T}\right)^T \mathbf{\Theta_T} \mathbf{h}_{TR}^{\mathcal{Z}_T}$ represents the effective cascaded channel gain for radar echo.

## III. Performance Analysis

### A. Statistical Characterization of Cascaded Channel Power

Let $h_{D_n,l}^{\mathcal{Z}_n} = |h_{D_n R_l}|\,|h_{R_l B}|$. Then $h_{D_n,l}^{\mathcal{Z}_n} \overset{\text{approx}}{\sim} \text{Gamma}\,(\alpha_n, \beta_n)$, where $\alpha_n = (\mathbb{E}[h_{D_n,l}^{\mathcal{Z}_n}])^2/\text{Var}[h_{D_n,l}^{\mathcal{Z}_n}]$ and $\beta_n = \mathbb{E}[h_{D_n,l}^{\mathcal{Z}_n}]/\text{Var}[h_{D_n,l}^{\mathcal{Z}_n}]$ are the shape and rate parameters, respectively. Mean and variance are computed using $k^{th}$ moment $\mathbb{E}\left[\left(h_{D_n,l}^{\mathcal{Z}_n}\right)^k\right] = \frac{\Gamma\left(m_{D_n R}+\frac{k}{2}\right)\Gamma\left(m_{RB}+\frac{k}{2}\right)}{\vartheta^k \Gamma(m_{D_n R})\,\Gamma(m_{RB})}$, for $k \geq 1$, where $\vartheta = \sqrt{m_{D_n R} m_{RB}/\left(\Omega_{D_n R}\Omega_{RB}\right)}$ [11, (71)] and they are given by (8) and (9), respectively. Then, $h_{D_n}^{\mathcal{Z}_n} = \sum_{l\in\mathcal{Z}_n} |h_{D_n,l}^{\mathcal{Z}_n}| \overset{\text{approx}}{\sim} \text{Gamma}\,(L_n\alpha_n, \beta_n)$ with PDF $f_{h_{D_n}^{\mathcal{Z}_n}}(x) = x^{L_n\alpha_n - 1} e^{-x\beta_n} \beta_n^{L_n\alpha_n}/\Gamma(L_n\alpha_n)$, for $x \geq 0$. Using [11, (18)], we get the cumulative distribution function (CDF) of $|h_{D_n}^{\mathcal{Z}_n}|^2$ as

$$F_{|h_{D_n}^{\mathcal{Z}_n}|^2}(x) = \frac{\gamma(L_n\alpha_n, \beta_n\sqrt{x})}{\Gamma(L_n\alpha_n)}, \quad x \geq 0, \tag{5}$$

where $\Gamma(.)$ and $\gamma(.,.)$ denote the Gamma function [15, (8.310)] and lower incomplete Gamma function [15, (8.350)], respectively. Similarly, the PDF and CDF of $|h_{D_f}^{\mathcal{Z}_f}|^2$ can be obtained owing to the identical reflection and propagation characteristics across RIS zones except that $m_{D_n R}$ and $\Omega_{D_n R}$ are replaced by $m_{D_f R}$ and $\Omega_{D_f R}$, respectively.

Assuming channel reciprocity, the effective channel gain of target echo corresponding to target-zone $\mathcal{Z}_T$ is $g_T^{\mathcal{Z}_T} = \left(\sum_{l\in\mathcal{Z}_T} |h_{TR_l}|\,|h_{R_l B}|\right)^2$. The CDF of the channel power gain $|g_T^{\mathcal{Z}_T}|^2$ can be shown to be

$$F_{|g_T^{\mathcal{Z}_T}|^2}(x) = \frac{\gamma(L_T\alpha_T, \beta_T x^{1/4})}{\Gamma(L_T\alpha_T)}, \quad x \geq 0, \tag{6}$$

where $\alpha_T$ and $\beta_T$ denote the shape and scale parameters of the Gamma-distributed RV $\left(\sum_{l\in\mathcal{Z}_T} |h_{TR_l}|\,|h_{R_l B}|\right)$. Similar to $\alpha_n$ and $\beta_n$, $\alpha_T$ and $\beta_T$ can be obtained by replacing $m_{D_n R}$ and $\Omega_{D_n R}$ with $m_{TR}$ and $\Omega_{TR}$, respectively.

**Lemma** *1:* The PDF of the channel power gain $|h_{D_n}|^2$, where $h_{D_n} = h_{D_n}^{\mathcal{Z}_n} + h_{D_n}^{\mathcal{Z}_f} + h_{D_n}^{\mathcal{Z}_T}$ be the effective channel from $D_n$ to the BS considering reflections from all three zones of the RIS, can be expressed as

$$f_{|h_{D_n}|^2}(x) = \frac{1}{\sigma_n^2} \exp\left(-\frac{x+\nu_n^2}{\sigma_n^2}\right) I_0\left(\frac{2\nu_n\sqrt{x}}{\sigma_n^2}\right), \; x \geq 0, \tag{7}$$

where $\nu_n = \mu_{\mathcal{Z}_n} = \mathbb{E}\left[h_{D_n}^{\mathcal{Z}_n}\right]$ and $\sigma_n^2 = \sigma_{n,\mathcal{Z}_n}^2 + \sigma_{n,\mathcal{Z}_f}^2 + \sigma_{n,\mathcal{Z}_T}^2$, and $I_0(\cdot)$ denotes the modified Bessel function of the first kind and order zero. The mean and the variance parameters, $\mu_{\mathcal{Z}_n}$ and $\sigma_{n,\mathcal{Z}_j}^2$, $j \in \{n, f, T\}$ are defined as

$$\mu_{\mathcal{Z}_n} = L_n \sqrt{\frac{\Omega_{D_n R}\,\Omega_{RB}}{\tilde{m}}} \left(\frac{\Gamma\left(m_{D_n R}+\frac{1}{2}\right)\Gamma\left(m_{RB}+\frac{1}{2}\right)}{\Gamma(m_{D_n R})\,\Gamma(m_{RB})}\right), \tag{8}$$

$$\sigma_{n,\mathcal{Z}_n}^2 = \Xi_n \left[1 - \frac{1}{\tilde{m}} \left(\frac{\Gamma\left(m_{D_n R}+\frac{1}{2}\right)\Gamma\left(m_{RB}+\frac{1}{2}\right)}{\Gamma(m_{D_n R})\,\Gamma(m_{RB})}\right)^2\right], \tag{9}$$

$$\sigma_{n,\mathcal{Z}_f}^2 = L_f\,\Omega_{D_n R}\,\Omega_{RB}, \quad \sigma_{n,\mathcal{Z}_T}^2 = L_T\,\Omega_{D_n R}\,\Omega_{RB}, \tag{10}$$

where $\Xi_n = L_n\,\Omega_{D_n R}\,\Omega_{RB}$ and $\tilde{m} = m_{D_n R}\,m_{RB}$.

*Proof:* We note that by central limit theorem (CLT), $h_{D_n}^{\mathcal{Z}_n}$ can be approximated as one-sided Gaussian RV, $h_{D_n}^{\mathcal{Z}_n} \sim \mathcal{N}^+(\mu_{\mathcal{Z}_n}, \sigma_{n,\mathcal{Z}_n}^2)$ with the normalization factor $\psi_{Dn} = 1/Q\left(-\mu_{\mathcal{Z}_n}/\sigma_{n,\mathcal{Z}_n}^2\right)$, where $Q(.)$ denotes Marcum Q-function, $h_{D_n}^{\mathcal{Z}_f} \sim \mathcal{CN}(0, \sigma_{n,\mathcal{Z}_f}^2)$, and $h_{D_n}^{\mathcal{Z}_T} \sim \mathcal{CN}(0, \sigma_{n,\mathcal{Z}_T}^2)$ represent the reflected channel components corresponding to the respective zones [8]. Hence, $h_{D_n} \sim \mathcal{CN}(\mu_n, \sigma_n^2)$, where $\mu_n = \mu_{\mathcal{Z}_n}$ and $\sigma_n^2 = \sigma_{n,\mathcal{Z}_n}^2 + \sigma_{n,\mathcal{Z}_f}^2 + \sigma_{n,\mathcal{Z}_T}^2$. Thus, $|h_{D_n}|^2$ is the squared magnitude of a nonzero-mean complex Gaussian and it follows a Rician-power [16], equivalently, a scaled non-central chi-squared distribution with the PDF given by (7). ■

Similarly, the PDF of $|h_{D_f}|^2$ can be obtained using (7), with $v_f = \mu_{\mathcal{Z}_f}$ and $\sigma_f^2 = \sigma_{f,\mathcal{Z}_n}^2 + \sigma_{f,\mathcal{Z}_f}^2 + \sigma_{f,\mathcal{Z}_T}^2$, where $\mu_{\mathcal{Z}_f}$ and $\sigma_{f,\mathcal{Z}_j}^2$, $j \in \{n, f, T\}$ are acquired by replacing $m_{D_n R}$ and $\Omega_{D_n R}$ by $m_{D_f R}$ and $\Omega_{D_f R}$, respectively in (8) - (10) .

The effective radar echo channel considering reflections from all zones of the RIS is given by $g_T = g_T^{\mathcal{Z}_n} + g_T^{\mathcal{Z}_f} + g_T^{\mathcal{Z}_T}$. By CLT, we can show that $g_T \sim \mathcal{CN}(\mu_T, \sigma_T^2)$, where $\mu_T = \mathbb{E}\left[g_T\right] = \mathbb{E}\left[g_T^{\mathcal{Z}_T}\right]$ denotes mean and $\sigma_T^2$ is the covariance of $g_T$, which is obtained numerically. We can show

that $\mathbb{E}\left[g_T^{\mathcal{Z}_n}\right] = \mathbb{E}\left[g_T^{\mathcal{Z}_f}\right] = 0$ and $\mathbb{E}\left[g_T^{\mathcal{Z}_T}\right] = L_T \Omega_{BR}\,\Omega_{RT} + L_T\,(L_T-1)\left(\frac{\Omega_{BR}\Omega_{RT}}{m_{BR}m_{RT}}\right)\left(\frac{\Gamma(m_{BR}+1/2)\Gamma(m_{RT}+1/2)}{\Gamma(m_{BR})\Gamma(m_{RT})}\right)^2$.

### B. Outage Probability of Communication and Sensing

We now analyze the outage probability of both communication and sensing. For analytical tractability, we consider the average radar echo interference power in the denominator of (2) and (3). The gap between instantaneous versus average echo interference power is negligible, as shall be verified in Fig. 2a. Note that $\mathbb{E}\left[|g_T|^2\right] = |\mu_T|^2 + \sigma_T^2$ since $\sigma_T^2 = \mathbb{E}\left[|g_T - \mu_T|^2\right]$. The received radar echo interference power $I_R = P_T \varrho |g_T|^2$, whose average value is

$$\mathbb{E}[I_R] = P_T \varrho \left(|\mu_T|^2 + \sigma_T^2\right). \tag{11}$$

*Outage Probability of Near User:* Under the SIC decoding order in uplink NOMA, the BS first detects the near-user signal. Let $\gamma_{th,c}$ denotes the minimum SINR threshold required for the successful decoding of the communication signals.

**Result** *1:* The outage probability of uplink near-user communication for the proposed RIS-ISAC-NOMA scheme is

$$P_{\text{out}}^{D_n} = e^{\frac{-\nu_n^2}{\sigma_n^2}} \int_0^\infty \frac{\gamma\left(L_n\alpha_n, \beta_n\sqrt{A\sigma_n^2 u + B}\right)}{\Gamma(L_n\alpha_n)} I_0\left(\frac{2\nu_n\sqrt{u}}{\sigma_n}\right) e^{-u}\, du, \tag{12}$$

where $A = \gamma_{th,c} P_f / P_n$ and $B = \gamma_{th,c}(\mathbb{E}[I_R] + \sigma_{\text{BS}}^2)/P_n$.

*Proof:* The outage probability for $D_n$ is $P_{\text{out}}^{D_n} = \Pr\{\gamma_{D_n} < \gamma_{th,c}\}$. From (2), it can be rewritten as $P_{\text{out}}^{D_n} = \Pr\left\{|h_{D_n}^{\mathcal{Z}_n}|^2 < A|h_{D_f}|^2 + B\right\} = \int_0^\infty F_{|h_{D_n}^{\mathcal{Z}_n}|^2}(Ax + B)\, f_{|h_{D_f}|^2}(x)\, dx\}$. Substituting the CDF of $|h_{D_n}^{\mathcal{Z}_n}|^2$ from (5) and the PDF of $|h_{D_f}|^2$from (7) and using the transformation $u = x/\sigma_n^2$, we get (12). ■

The integral in (12) can not be simplified further to the best of our knowledge. However, using Gauss–Laguerre quadrature [17, (25.4.45)], it can be approximated as $P_{\text{out}}^{D_n} \approx \sum_{i=1}^N w_i A_i$, where $A_i = e^{-\frac{\nu_n^2}{\sigma_n^2}} \frac{\gamma\left(L_n\alpha_n, \beta_n\sqrt{A\sigma_n^2 x_i + B}\right)}{\Gamma(L_n\alpha_n)} I_0\left(\frac{2\nu_n\sqrt{x_i}}{\sigma_n}\right)$ with $x_i$ and $w_i$, for $1 \le i \le N$ , are $N$ Gauss Laguerre abscissas and weights, respectively.

*Outage Probability of Far User:* The far-user signal is decoded after canceling the near-user interference from the composite received signal at the BS. Let $\gamma_{SIC}$ denotes the minimum SINR threshold required for the successful decoding of near-user signal in the SIC process. The outage probability for $D_f$ is expressed as

$$P_{\text{out}}^{D_f} = 1 - \Pr\{\gamma_{D_n} > \gamma_{SIC},\, \gamma_{D_f} > \gamma_{th,c}\}. \tag{13}$$

Since the two events are statistically independent,

$$\begin{aligned} P_{\text{out}}^{D_f} &= 1 - \Pr\{\gamma_{D_n} > \gamma_{SIC}\}\, \Pr\{\gamma_{D_f} > \gamma_{th,c}\}, \\ &\triangleq 1 - (1 - P_{out,SIC}^{D_n})(1 - P_{out,th,c}^{D_f}). \end{aligned} \tag{14}$$

Now, by replacing $\gamma_{th,c}$ with $\gamma_{SIC}$, $P_{out,SIC}^{D_n}$ is obtained from the near-user outage probability analysis in (12). Let $A' = \gamma_{th,c}\delta_n P_n / P_f$, $B' = \gamma_{th,c}(\mathbb{E}[I_R] + \sigma_{\text{BS}}^2)/P_f$. Then, the term $P_{out,th,c}^{D_f} = \int_0^\infty F_{|h_{D_f}^{\mathcal{Z}_f}|^2}(A'q + B')\, f_{|h_{D_n}^2|}(q)\, dq$. It can be simplified as

$$P_{out,th,c}^{D_f} = e^{-\frac{\nu_f^2}{\sigma_f^2}} \int_0^\infty \frac{\gamma\left(L_f\alpha_f, \beta_f\sqrt{A'\sigma_f^2 u + B'}\right)}{\Gamma(L_f\alpha_f)} I_0\left(\frac{2\nu_f\sqrt{u}}{\sigma_f}\right) e^{-u} du. \tag{15}$$

Using Gauss Laguerre quadrature, $P_{out,th,c}^{D_f} \approx \sum_{j=1}^N w_j B_j$. Here, $B_j = \exp\left(-\frac{\nu_f^2}{\sigma_f^2}\right) \frac{\gamma\left(L_f\alpha_f, \beta_f\sqrt{A'\sigma_f^2 x_j + B'}\right)}{\Gamma(L_f\alpha_f)} I_0\left(\frac{2\nu_f\sqrt{x_j}}{\sigma_f}\right)$. Substituting this in (14), we obtain the exact expression for $P_{\text{out}}^{D_f}$ in the form of single-integrals and its approximation in closed-form summations.

*Outage Probability of Radar Echo:* After successful decoding of communication signals, the BS finally decodes the radar echo reflected from $T$. The corresponding outage probability $P_{\text{out}}^{BS}$ with the minimum SINR threshold $\gamma_{th,r}$ for successful decoding radar echo is given by

$$\begin{aligned} P_{\text{out}}^{BS} &= 1 - \Pr\{\gamma_{D_n} > \gamma_{SIC}\}\Pr\{\gamma_{D_f} > \gamma_{SIC}\}\Pr\{\gamma_{BS} > \gamma_{th,r}\} \\ &\triangleq 1 - (1 - P_{out,SIC}^{D_n})(1 - P_{out,SIC}^{D_f})(1 - P_{out,th,r}^{BS}). \end{aligned} \tag{16}$$

Using (4), we can write $P_{out,th,r}^{BS} = \Pr\{\gamma_{BS} \le \gamma_{th,r}\} = \int_0^\infty \int_0^\infty F_{|g_T^{\mathcal{Z}_T}|^2}(c_n x + c_f y + c_T) f_{|h_{D_n}|^2}(x) f_{|h_{D_f}|^2}(y)\, dx\, dy$, where $c_n = \gamma_{th,r}\delta_n P_n / P_T$, $c_f = \gamma_{th,r}\delta_f P_f / P_T$, and $c_T = \gamma_{th,r}\sigma_{BS}^2 / P_T$. Using the CDF (c.f. (6)) and PDF (c.f. (7)) and applying Gauss Laguerre quadrature, $P_{out,th,r}^{BS}$ simplifies to the double summation term in (17).

**Result** *2:* The outage probability of decoding the radar echo reflected from $T$ is given by

$$P_{\text{out}}^{BS} \approx 1 - \left[1 - \sum_{i=1}^N w_i A_i\right]\left[1 - \sum_{j=1}^N w_j B_j\right]\left[1 - \sum_{i=1}^N \sum_{j=1}^N u_i v_j\, C_{ij}\right], \tag{17}$$

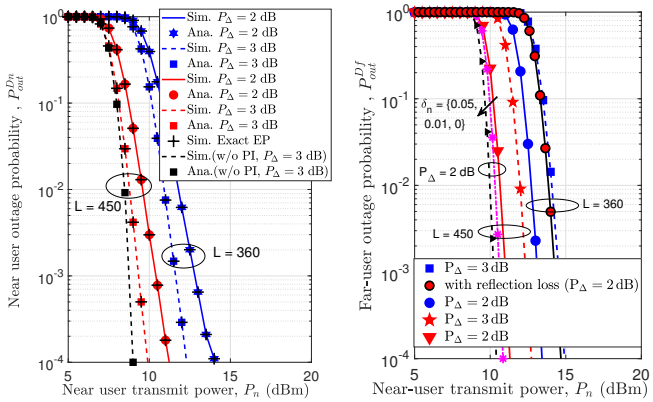
where $C_{ij} = \frac{1}{\Gamma(L_T\alpha_T)} \gamma\left(L_T\alpha_T, \beta_T\left(c_n\sigma_n^2 x_i + c_f\sigma_f^2 y_j + c_T\right)^{1/4}\right) \times I_0\left(\frac{2\nu_n\sqrt{x_i}}{\sigma_n}\right) I_0\left(\frac{2\nu_f\sqrt{y_j}}{\sigma_f}\right) e^{-\left(\frac{\nu_n^2}{\sigma_n^2} + \frac{\nu_f^2}{\sigma_f^2}\right)}$ and $(u_i, x_i)$, $(v_j, y_j)$, $1 \le i, j \le N$ are $N$ Gauss Laguerre abscissas and weights, respectively. We have found that $N = 25$ terms suffice for the considered parameters, yielding a computationally efficient closed-form outage approximation.

### C. Average Sensing Information Rate (ASIR)

The average sensing information rate $R_{\text{avg}}$ characterizes the long-term sensing throughput achieved for radar echo estimation at the BS [14]. It is defined as $R_{\text{avg}} = \mathbb{E}\left[\frac{\varpi}{2T}\log_2(1 + 2TB\,\gamma_{BS})\right]$, where $\varpi$ is radar duty cycle, $T$ is radar pulse duration, and $B$ is the bandwidth. It can be written in terms of the complementary CDF of $\gamma_{BS}$ as $R_{\text{avg}} = \frac{\varpi}{2T\ln 2}\int_0^\infty \frac{\Pr\{\gamma_{BS} > x/a_2\}}{1+x}\, dx$, where $a_2 = 2TB$ and

$$R_{\text{avg}} \approx \frac{\varpi}{2T\ln 2}\int_0^\infty \frac{1}{1+x}\left[1 - \sum_{i=1}^N \sum_{j=1}^N u_i v_j\, C_{ij}\right] dx, \tag{18}$$

where $C_{ij}$ is given after (17), in which $\gamma_{th,r} = x/a_2$ is used to evaluate other constant parameters.




(a) Outage probability for $D_n$ (b) Outage probability for $D_f$

Fig. 2: Outage probability performance for $D_n$ and $D_f$ under varying REs $L = \{360, 450\}$, power-backoff level $P_\Delta = \{2, 3\}$ dB, and impSIC factor $\delta_n = \{0.05, 0.01, 0\}$ with $\gamma_{th,c} = 1$ and $P_T = 20$ dBm; Marker shows analysis, line shows simulations (b).

## IV. Numerical Results

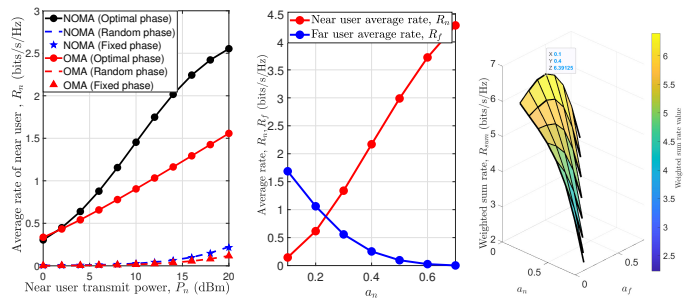
Monte Carlo simulation results are presented in this section to validate the analytical derivations and provide further insights into the proposed system performance. The parameters used in the simulation are listed in Table II.

TABLE II: Simulation Parameters

| Simulation Parameters | Values |
|---|---|
| Bandwidth ($B$) | 10 MHz |
| Nakagami shape parameter ($m$) | 3 |
| Carrier Frequency ($f_c$) | 5.8 GHz |
| Path loss exponent ($\alpha_c, \alpha_r$) | (2.5, 4) |
| Noise Power ($\sigma_{BS}^2$) | $-104$ dBm |
| Radar Cross Section ($\sigma_{\text{RCS}}$) | 50 m$^2$ [18] |
| Variance of Time Delay Fluctuation ($\sigma_\tau^2$) | $10^{-6}$ [19] |
| Power-Backoff Factor ($P_\Delta$) | [2, 3] dB |
| Near User Transmit Power ($P_n$) | 0–20 dBm |
| BS Sensing Signal Power ($P_T$) | 10–40 dBm |
| RIS Position | (100, 50) m |

Fig. 2 illustrates the outage probability of the near and far user versus near user transmit power $P_n$ considering equal partitioning coefficients for RIS. Simulation results match the analysis well. From Fig. 2a, we see that increasing the number of REs ($L$) from 360 to 450 significantly reduces the outage probability. Furthermore, a higher power-backoff at the far user ($P_\Delta = 3$ dB) reduces interference to decode near user's signal, leading to a lower outage probability for $D_n$. Furthermore, we see that ignoring the inter-partitioning interference (w/o PI) reduces the outage probability by decreasing the effective interference power, as depicted for $P_\Delta = 3$ dB, $L = 450$. We observe that replacing the instantaneous echo interference power (Exact EP) with its statistical average introduces no visible impact.

In Fig. 2b, the outage performance of the far user is plotted. As $P_n$ increases, the outage probability of the far user decreases since $P_f$ is linearly related to $P_n$ ($P_f = 10^{-P_\Delta/10} P_n$). For a given $P_n$, as $P_\Delta$ increases from 2 dB to 3 dB, the outage probability for the far user increases due to lower




(a) $R_n$ vs $P_n$ (b) $R_n$ and $R_f$ vs $a_n$ (c) Weighted sum rate

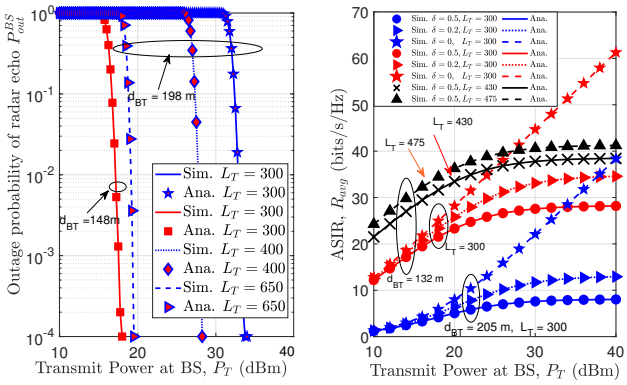
Fig. 3: (a) Average rate for near user, (b) Average rate of both users vs $a_n$ under fixed $a_T$ and (c) Weighted sum rate of sensing and communication vs $a_n$ and $a_f$ with $L = 400$, $w_c = 0.9$, $w_r = 0.1$, $P_T = 20$ dBm, $P_n = 10$ dBm and $P_\Delta = 5$dB.

transmit power $P_f$. As the impSIC factor $\delta_n$ decreases from 0.05 to 0.01, outage probability of $D_f$ decreases. This is because the residual interference from $D_n$ after SIC decreases and approaching the ideal SIC case ($\delta_n = 0$). Increasing the number of RIS REs improves the performance for the far user. For $P_\Delta = 2$ dB and $L = 360$, we consider RIS reflection loss, in which the reflection amplitude coefficient depends on the phase configured [20, (5)]. We see that the outage probability increases due to reduced RIS reflected signal power.

Fig. 3a compares the average rate $R_n \triangleq \mathbb{E}[\log_2(1 + \gamma_{D_n})]$ of the near-user for the proposed NOMA scheme with conventional OMA scheme. Three RIS phase configuration schemes are considered under equal partitioning: (i) optimal phase $\theta_{l_n} = -(\theta_{D_n R_{l_n}} + \theta_{R_{l_n} B})$, (ii) random phase, where the RIS phases are chosen randomly from an uniform distribution, and (iii) fixed phase, where a fixed value for $\theta_{l_n}$, for $l_n \in \mathcal{Z}_n$ is used. NOMA outperforms conventional OMA, demonstrating more efficient resource utilization for NOMA. With optimal phase shifts, NOMA achieves the highest average rate compared to random and fixed phase configurations. The average rate $R_f = \mathbb{E}[\log_2(1 + \gamma_{D_f})]$ for the far-user also exhibits similar behavior (not shown for brevity).

Fig. 3b illustrates the variation of the achievable rates $R_n$ and $R_f$ as a function of RIS partitioning coefficient of near user $a_n$, for fixed $a_T = 0.25$. Thus, $a_f = 1 - a_n - a_T$ is varied. It is observed that as $a_n$ increases, $R_n$ increases but $R_f$ decreases due to reduced REs allocated for $D_f$. Moreover, $a_n = 0.22$ and $a_f = 0.55$ provides almost equal average rate to both the users. Thus, proper choice of partitioning coefficients is crucial to ensure fairness between the users.

Fig. 3c illustrates the impact of partitioning coefficients $(a_n, a_f)$ variation on weighted sum rate of the ISAC system, which is defined as $R_{\text{sum}} = w_c(R_n + R_f) + w_r R_{\text{avg}}$, where $w_c$ and $w_r$ denote the weights associated with communication and sensing functions, respectively. The fraction of REs allocated for sensing is set as $a_T = 1 - a_n - a_f$. We see that maximum value $R_{sum} = 6.4$ bits/s/Hz is obtained with $a_n = 0.1$ and $a_f = 0.4$. This indicates that the sensing task requires 50% of the available REs and then the far user is




(a) Sensing outage probability (b) ASIR

Fig. 4: (a) Impact of target position and REs allocated on $P_{\text{out}}^{\text{BS}}$ with $\delta_n = \delta_f = 0.2$, (b) Impact of impSIC factors on ASIR ($P_n = [0, 20]$ dBm, $P_\Delta = 5$ dB, and $[L_n, L_f] = [90, 140]$).

prioritized in allocating REs to ensure fairness among sensing and communication tasks based on the choice of $w_c$ and $w_r$.

Fig. 4a shows the radar echo outage probability $P_{\text{out}}^{\text{BS}}$ versus BS transmit power $P_T$ for two positions of the target, when $D_n$ and $D_f$ are at 598 m and 748 m, respectively, and the RIS is at 100 m from the BS. For a nearby target ($d_{BT} = 148$ m), $P_{\text{out}}^{\text{BS}}$ decreases rapidly with $P_T$ due to stronger received echo power, while a farther target ($d_{BT} = 198$ m) exhibits higher outage probability due to larger path loss at fixed $L_T = 300$. However, increasing $L_T$ to 450 and 650 significantly reduces $P_{\text{out}}^{\text{BS}}$, illustrating the efficacy of adaptive RE allocation.

Fig. 4b shows the ASIR versus $P_T$ for two target positions. For $d_{BT} = 205$ m and $L_T = 300$, decreasing the impSIC factor ($\delta_n = \delta_f = \delta$) increases the ASIR due to reduced residual interference from the communication users' signals. As $P_T$ increases, the ASIR initially increases and then saturates for $\delta > 0$, while it increases monotonically for $\delta = 0$ (perfect SIC). A nearer target ($d_{BT} = 132$ m) achieves higher ASIR due to stronger echoes and lower path loss. The performance degradation from impSIC can be mitigated by increasing $L_T$; for $d_{BT} = 132$ m and $\delta = 0.5$, higher $L_T$ yields larger ASIR.

## V. Conclusion

We investigated an RIS-assisted ISAC-NOMA uplink system considering imperfect SIC. We derived novel outage probability expressions for both communication and sensing and also analyzed an ASIR metric for sensing. Under RIS partitioning, inter-zone interference at BS was modeled to capture RIS reflection coupling effects. Simulation results demonstrated that increasing no. of REs for sensing improved sensing performance. The impact of power-backoff, target distance, impSIC factor, no. of RIS elements, and partitioning coefficients selection on outage probability and average rate were studied. Comparisons with OMA and different phase configurations confirmed the advantages of NOMA.

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## References

[1] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 6, pp. 1728–1767, Mar. 2022.

[2] S. Aldirmaz-Colak, M. Namdar, A. Basgumus, S. Özyurt, S. Kulac, N. Calik, M. Akif Yazici, A. Serbes, and L. Durak-Ata, "A comprehensive review on isac for 6G: Enabling technologies, security, and AI/ML perspectives," *IEEE Access*, vol. 13, pp. 97 152–97 193, Jun. 2025.

[3] Z. Wang, Y. Liu, X. Mu, Z. Ding, and O. A. Dobre, "NOMA empowered integrated sensing and communication," *IEEE Commun. Lett.*, vol. 26, no. 3, pp. 677–681, Mar. 2022.

[4] J. Zuo and Y. Liu, "Reconfigurable intelligent surface assisted NOMA empowered integrated sensing and communication," in *Proc. Globecom*, Jan. 2022, pp. 1028–1033.

[5] X. Gan, C. Huang, Z. Yang, X. Chen, F. Bader, Z. Zhang, C. Yuen, Y. Liang Guan, and M. Debbah, "Modeling and coverage analysis of RIS-assisted integrated sensing and communication networks," *IEEE Trans. Wireless Commun.*, vol. 24, no. 7, pp. 6006–6019, Jul. 2025.

[6] J. Zuo, Y. Liu, C. Zhu, Y. Zou, D. Zhang, and N. Al-Dhahir, "Exploiting NOMA and RIS in integrated sensing and communication," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 12 941–12 955, Oct. 2023.

[7] W. Lyu, Y. Xiu, X. Li, S. Yang, P. L. Yeoh, Y. Li, and Z. Zhang, "Hybrid NOMA assisted integrated sensing and communication via RIS," *IEEE Trans. Veh. Technol.*, vol. 73, no. 5, pp. 7368–7373, May 2024.

[8] H. Raj and B. R. Manoj, "Cascaded channel statistics and target detection analysis for IRS-assisted ISAC employing NOMA," in *Proc. Nat. Conf. Commun.*, May 2025, pp. 1–6.

[9] A. S. Parihar, K. Singh, C.-P. Li, V. Bhatia, and T. Q. Duong, "On the performance of NOMA-enabled multi-sector RIS-aided with ISAC," in *Proc. ICC Workshop*, Sep 2025, pp. 1592–1597.

[10] S. Mondal, K. Singh, D. W. Kwan Ng, C.-P. Li, and Z. Ding, "Outage performance of RIS-aided NOMA ISAC network for LEO satellite system," in *Proc. ICC Workshop*, Sep. 2025, pp. 1556–1561.

[11] T. N. Do, G. Kaddoum, T. L. Nguyen, D. B. da Costa, and Z. J. Haas, "Multi-RIS-aided wireless systems: Statistical characterization and performance analysis," *IEEE Trans. Commun.*, vol. 69, no. 12, pp. 8641–8658, Dec. 2021.

[12] S. Hashemi, H. Jiang, and M. Ardakani, "Optimal configuration of reconfigurable intelligent surfaces with arbitrary discrete phase shifts," *IEEE Trans. Commun.*, vol. 72, no. 12, pp. 8047–8060, Dec. 2024.

[13] N. P. Le, L. C. Tran, X. Huang, J. Choi, E. Dutkiewicz, S. L. Phung, and A. Bouzerdoum, "Performance analysis of uplink NOMA systems with hardware impairments and delay constraints over composite fading channels," *IEEE Trans. Veh. Technol.*, vol. 70, no. 7, pp. 6881–6897, Jul. 2021.

[14] C. Zhang, W. Yi, Y. Liu, and L. Hanzo, "Semi-integrated-sensing-and-communication (semi-ISAC): From OMA to NOMA," *IEEE Trans. Commun.*, vol. 71, no. 4, pp. 1878–1893, Feb. 2023.

[15] I. S. Gradshteyn and I. M. Ryzhik, *Table of Integrals, Series and Products*, 4th ed. Academic Press, 1980.

[16] M. Simon and M.-S. Alouini, *Digital Communication over Fading Channels*, 2nd ed. Wiley-Interscience, 2005.

[17] M. Abramowitz and I. Stegun, *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*, 9th ed. Dover, 1972.

[18] L. Sun, Z. Zhao, S. Wang, Z. Ding, and M. Peng, "On the study of non-orthogonal multiple access (NOMA)-assisted integrated sensing and communication (ISAC)," vol. 72, no. 11, pp. 7278–7293, Nov. 2024.

[19] A. R. Chiriyath, B. Paul, G. M. Jacyna, and D. W. Bliss, "Inner bounds on performance of radar and communications co-existence," *IEEE Trans. Signal Process.*, vol. 64, no. 2, pp. 464–474, Sep. 2016.

[20] S. Abeywickrama, R. Zhang, Q. Wu, and C. Yuen, "Intelligent reflecting surface: Practical phase shift model and beamforming optimization," *IEEE Trans. Commun.*, vol. 68, no. 9, pp. 5849–5863, Sep. 2020.